Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: M.Tech (CSE)& III
Name ofthePaper:Artificial Intelligence & Expert System
CourseCode: MCS 303
Time: 1.5 Hour Maximum Marks: 50

Note

(i) All Questions are compulsory.
(ii) Each question carries 10 marks.

Q.1 **(10 Marks)** **CO1**

(a) What are the core differences between Artificial Intelligence (AI) and Natural Intelligence? Provide examples.

OR

(b) Explain the underlying assumptions made in AI research. Why are these assumptions important?

Q.2 **(10 Marks)** **CO1**

(a) Define problem spaces in AI. How are they utilized in the process of problem-solving?

OR

(b) Explain Depth-First Search (DFS) and its application in AI. Compare its efficiency with BFS.

Q.3 **(10 Marks)** **CO1**

(a) Discuss the role of heuristic functions in AI. How do they enhance the efficiency of search algorithms?

OR

(b) What are the fundamental concepts of knowledge representation in AI? How does it facilitate problem-solving?

Q.4 **(10 Marks)** **CO2**

(a) Discuss the role of predicate logic in knowledge representation. Provide an example to illustrate its application.

OR

(b) Describe the concept of structured representation in AI. Why is it important?

Q.5 **(10 Marks)** **CO2**

(a) Compare and contrast various representations and mappings used in AI.

OR

(b) Discuss how semantic networks are constructed and utilized in AI for knowledge mapping.

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